

Transformations of Absolute-Value Functions

Answer Key

Algebra 1

Quick Answers

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|--------------|-----------------------------|
| 1. 4, 1 | 7. -2, -7, 2 |
| 2. -3, -5 | 8. -1, 7, -3, 9 |
| 3. 1, 6 | 9. $a = 3$, $k = -2$ |
| 4. -3, -1, 5 | 10. -2, 1, 7 |
| 5. -5, 3 | 11. $a = -2$, $k = 8$ |
| 6. 4, -4, 4 | 12. 1, 1, 5, $2 t - 3 + 1$ |
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Curriculum

Level: Algebra 1

HSA-CED.A.1–A.4 – *problems 9, 11, 12*

HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 10*

Difficulty 1–5 of 5 across 26 tagged checks.

Common wrong answers

3. *If they got 2: read the minus inside the bars as a shift LEFT and evaluated at $x = -1$*
7. *If they got -4: dropped the factor 3, so no vertical stretch was applied*
10. *If they got 2: reflected the graph but forgot the vertical stretch by 2*
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- 1.** Give the vertex of $g(x) = |x - 4| + 1$, and state the equation of its axis of symmetry.

Solution. The vertex sits where the expression inside the bars is zero, because $|\cdot|$ is smallest there. Solve $x - 4 = 0$, so $x = 4$. Then $g(4) = |4 - 4| + 1 = 1$. The axis of symmetry is the vertical line through the vertex, $x = 4$.

(4, 1)

- 2.** Describe in words how the graph of $g(x) = |x + 3| - 5$ is obtained from the graph of $f(x) = |x|$, then give the vertex of g .

Solution. Write $|x + 3|$ as $|x - (-3)|$: the inside is zero at $x = -3$, so the parent V slides 3 units LEFT. The -5 outside the bars lowers every output, a shift 5 units DOWN. Vertex: solve $x + 3 = 0$ to get $x = -3$, then $g(-3) = -5$.

(-3, -5)

- 3.** For $h(x) = -2|x - 1| + 6$: give the vertex, say whether the V opens up or down, and say whether the graph is narrower or wider than $y = |x|$.

Solution. Inside zero: $x - 1 = 0$, so $x = 1$, and $h(1) = -2|1 - 1| + 6 = 6$. Since $a = -2 < 0$ the V is reflected and opens DOWN; since $|a| = 2 > 1$ each arm rises twice as fast, so the graph is NARROWER than $y = |x|$.

$$(1, 6)$$

4. Sketch $g(x) = |x - 2| - 3$. Label the vertex and both x -intercepts.

Solution. Vertex: $x - 2 = 0$ gives $x = 2$ and $g(2) = -3$. For the x -intercepts set $g(x) = 0$: $|x - 2| = 3$, so $x - 2 = 3$ or $x - 2 = -3$, giving $x = 5$ and $x = -1$. Since $a = 1 > 0$ the V opens up with arms of slope ± 1 ; plot $(2, -3)$, then $(-1, 0)$ and $(5, 0)$, and draw the two rays.

$$(2, -3); x = -1 \text{ or } 5$$

5. The graph of $f(x) = |x + 1| - 4$ is a V with its vertex 4 units below the x -axis. Without graphing, find every x for which $f(x) = 0$.

Solution. The vertex is at $x = -1$, $y = -4$, and the V opens up, so it crosses the x -axis exactly twice — once on each arm. Set $|x + 1| - 4 = 0$, so $|x + 1| = 4$ and $x + 1 = \pm 4$. Then $x = 3$ or $x = -5$.

$$x = -5 \text{ or } x = 3$$

6. Sketch $g(x) = -|x| + 4$. Label the vertex and both x -intercepts.

Solution. There is no horizontal shift, so the vertex sits at $x = 0$ with $g(0) = 4$. The coefficient $a = -1$ reflects the parent V, so it opens down with arms of slope ± 1 . Setting $-|x| + 4 = 0$ gives $|x| = 4$, so $x = -4$ or $x = 4$.

$$(0, 4); x = -4 \text{ or } 4$$

7. Let $g(x) = 3|x + 2| - 7$. Give the vertex, the equation of the axis of symmetry, and the value of $g(1)$.

Solution. Inside zero: $x + 2 = 0$, so $x = -2$ and $g(-2) = -7$; the axis of symmetry is $x = -2$. Then $g(1) = 3|1 + 2| - 7 = 3(3) - 7 = 2$. Note the stretch is applied to $|x + 2|$ BEFORE the -7 , so the 3 multiplies only the distance term.

$$(-2, -7); g(1) = 2$$

8. The graph of $h(x) = 2|x - 3| - 8$ is a V opening up with vertex $(3, -8)$. Find every x with $h(x) = 0$, and every x with $h(x) = 4$.

Solution. A horizontal line above the vertex meets the V twice, symmetrically about $x = 3$. For $h(x) = 0$: $2|x - 3| = 8$, so $|x - 3| = 4$ and $x = 3 \pm 4$, giving $x = -1$ and $x = 7$. For $h(x) = 4$: $2|x - 3| = 12$, so $|x - 3| = 6$ and $x = 3 \pm 6$, giving $x = -3$ and $x = 9$. Each pair is the same distance left and right of the axis of symmetry.

$$x = -1, 7 \text{ and } x = -3, 9$$

9. An absolute-value function $g(x) = a|x - h| + k$ has vertex $(5, -2)$ and passes through $(7, 4)$. Find a and k , then write $g(x)$ in vertex form.

Solution. The vertex gives $h = 5$ and $k = -2$ directly. Substituting the point: $4 = a|7 - 5| + k$, that is $2a + k = 4$. With $k = -2$ this is $2a - 2 = 4$, so $2a = 6$ and $a = 3$. Since $a > 0$ the V opens up and is three times as steep as the parent.

$$a = 3, k = -2, g(x) = 3|x - 5| - 2$$

10. Sketch $g(x) = -2|x - 4| + 6$. Label the vertex, both x -intercepts, and the y -intercept, then list the transformations.

Solution. Vertex: $x - 4 = 0$ gives $x = 4$ and $g(4) = 6$. The y -intercept is $g(0) = -2|0 - 4| + 6 = -8 + 6 = -2$. The x -intercepts: $-2|x - 4| + 6 = 0$ gives $|x - 4| = 3$, so $x = 1$ and $x = 7$. Transformations of $y = |x|$: shift right 4, stretch vertically by a factor of 2, reflect across the x -axis, shift up 6.

$$(4, 6); g(0) = -2; x = 1 \text{ or } 7$$

11. The graph of an absolute-value function has vertex $(-3, 8)$ and passes through $(1, 0)$. Write the function in vertex form.

Solution. From the vertex, $h = -3$ and $k = 8$, so $g(x) = a|x + 3| + 8$. Substituting $(1, 0)$: $0 = a|1 + 3| + 8$, that is $4a + 8 = 0$, so $a = -2$. The negative a is the algebraic signal that the V opens downward, which it must, since the given point lies below the vertex.

$$g(x) = -2|x + 3| + 8$$

12. A drone's distance from the trailhead, in kilometres, after t minutes is $d(t) = |2t - 6| + 1$. (a) Rewrite in vertex form and give the vertex. (b) Explain what the vertex means. (c) Find every t at which the drone is 5 kilometres from the trailhead.

Solution. (a) Factor the 2 out of the bars: $|2t - 6| = |2(t - 3)| = 2|t - 3|$, so $d(t) = 2|t - 3| + 1$. The vertex is at $t = 3$ with $d(3) = 1$. (b) At $t = 3$ minutes the drone is at its closest approach to the trailhead, 1 kilometre away; before that it is coming in and after that it is going back out. (c) Set $2|t - 3| + 1 = 5$: then $|t - 3| = 2$, so $t = 3 \pm 2$, giving $t = 1$ and $t = 5$ minutes. Both times are valid because the domain of the trip includes them.

$$d(t) = 2|t - 3| + 1; V(3, 1); t = 1 \text{ or } 5$$